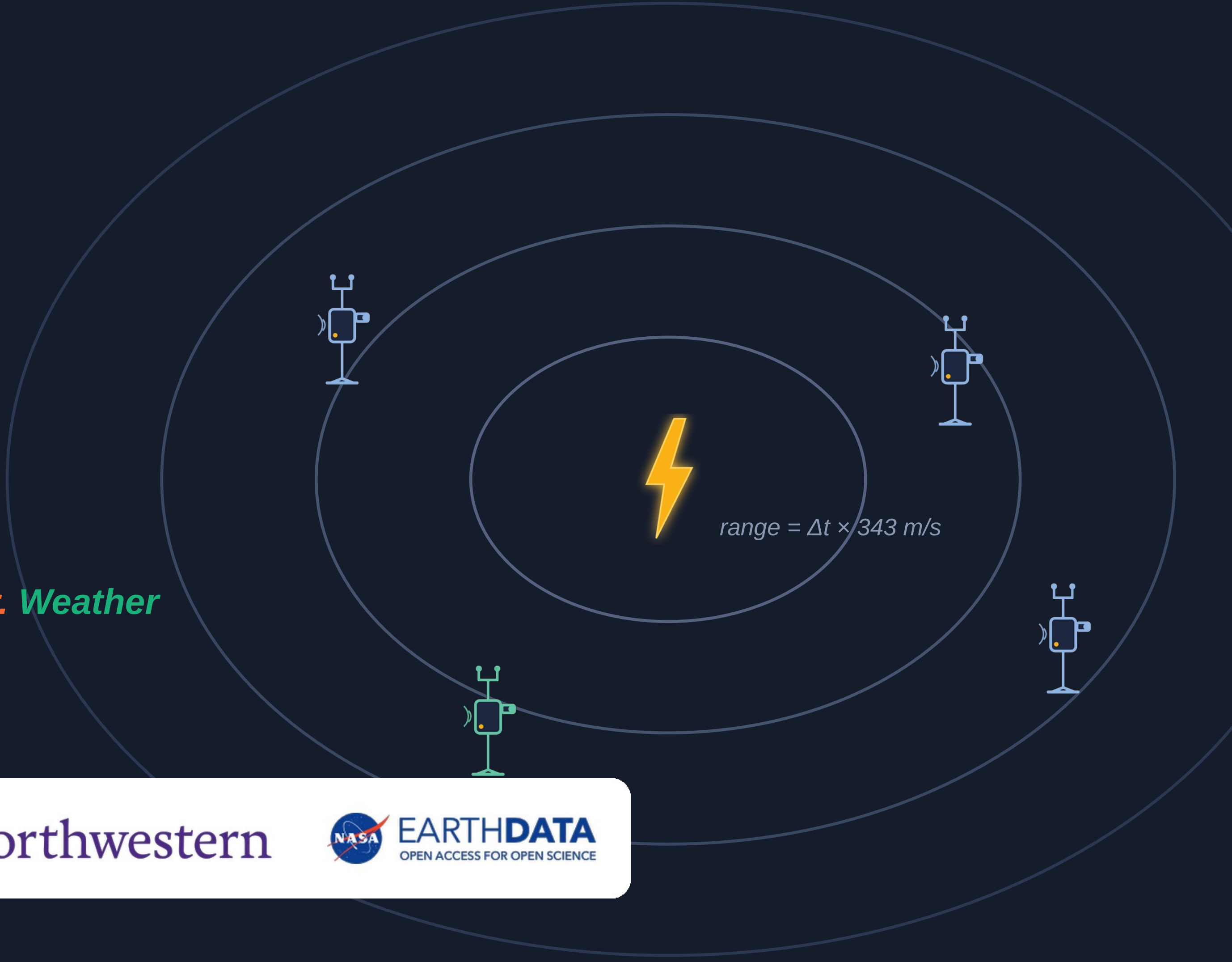
Sage FlashPoint

Multi-node lightning localization & wildfire ignition watch on the Sage fleet

Cameras catch the flash. Microphones catch the thunder. Weather decides when to listen.



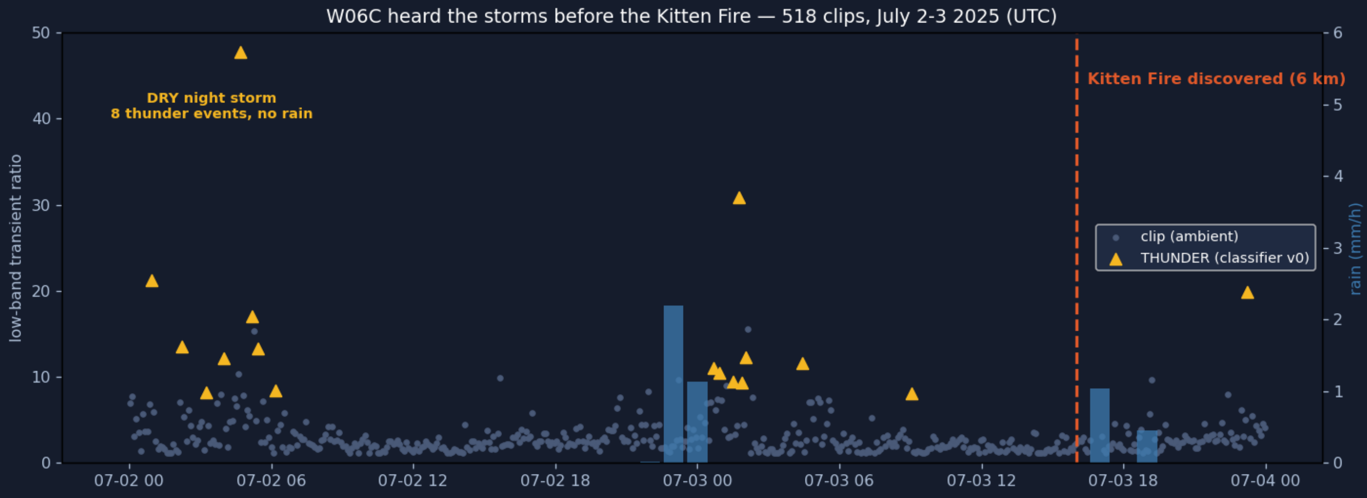
Samuel Watson · University of Hawaiʻi at Mānoa · Sage Summer Camp 2026 hackathon

This work was supported in part by the National Science Foundation under Awards No. 2331263 and 2436842.

THE CASE

A Sage node heard the fire start. Nobody was listening.

- Kitten Fire — Grand Teton WY, natural cause, discovered midday July 3 2025, six kilometers from node W06C.
- The node recorded straight through both suspect storms: audio snapshots, PTZ (pan-tilt-zoom) camera frames, full weather telemetry.
- Holdover fires smolder for hours or days before flaring. The strike is the earliest possible warning — and today nothing connects strikes to a follow-up watch.



Across 518 classified clips: a DRY night storm (0.0 mm rain) and a second storm the next evening — then the discovery line.

6 km
node to fire point

779
audio clips, July 1–4

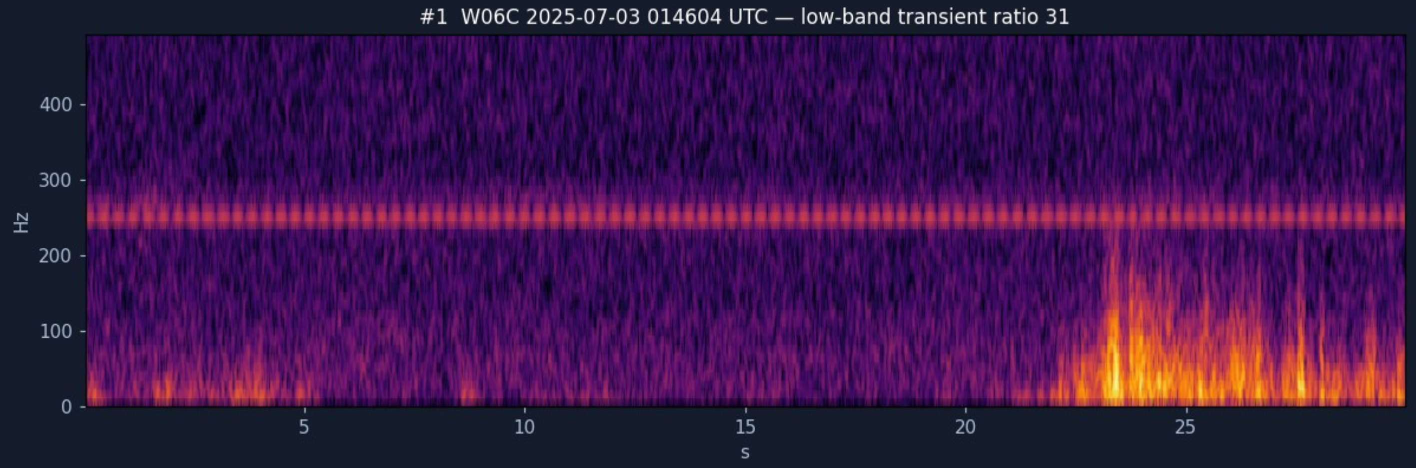
51k
weather records

THE FALSIFICATION

17 confident detections. Two satellites. Zero survivors.

0 / 17

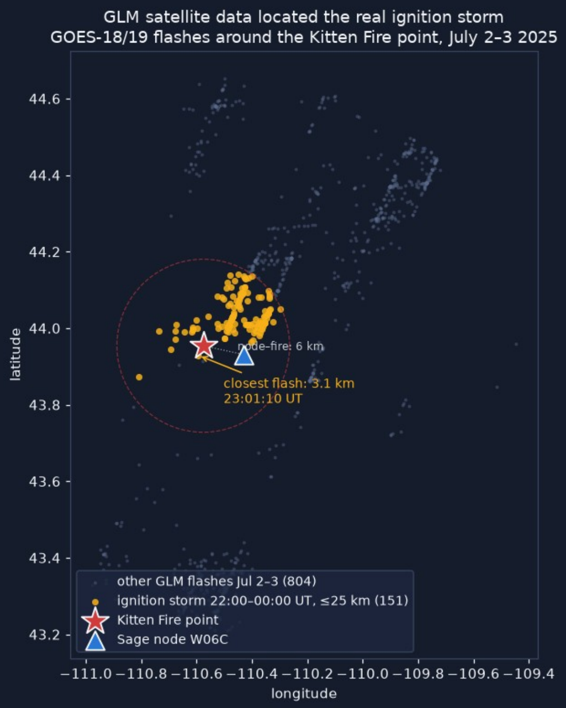
- Audio-only detector v0 flagged 17 thunder events across the storm nights — all confident, none wind-vetoed.
- Cross-checked against BOTH GOES-18 and GOES-19 (Geostationary Operational Environmental Satellites) lightning sensors — ±90-second window: not one flash within 50 km of any of them.
- **Single-modality detection false-alarms confidently.**
- That result became the architecture: standalone detections are nominations — never confirmations.



The strongest candidate of the 17 — low-band transient ratio 31 — of undetermined origin per two independent satellites.

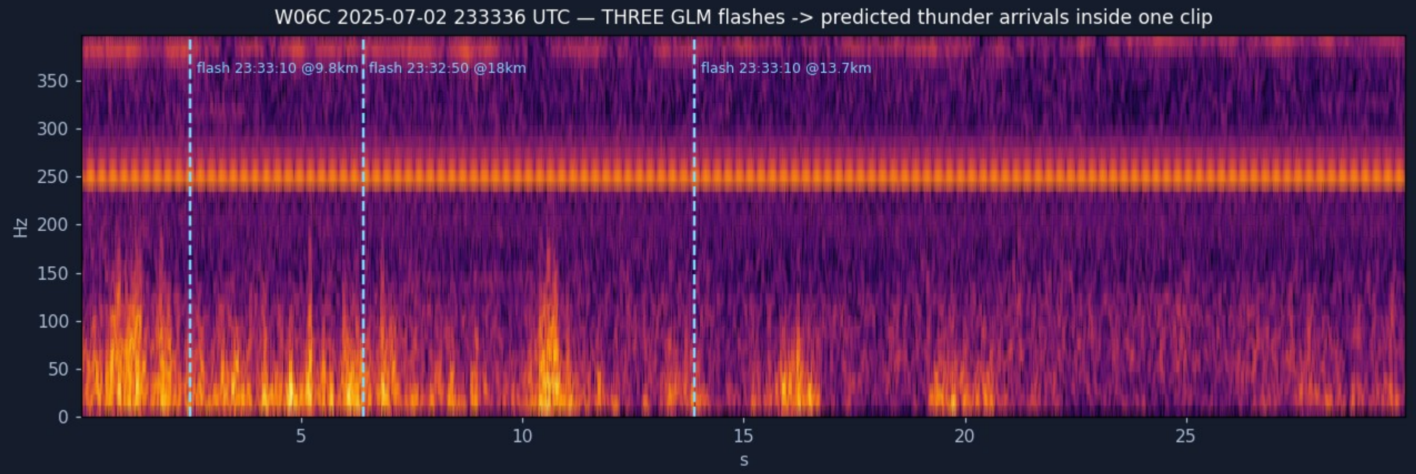
THE RECOVERY

Anchor on the flash, and the buried thunder appears



The real ignition storm, located by satellite scan: closest flash 3.1 km from the fire point, ~3 h before discovery. Data: NOAA (National Oceanic and Atmospheric Administration) GLM (Geostationary Lightning Mapper) via AWS (Amazon Web Services) Open Data.

- Each satellite flash gives a time-zero. Sound needs ~3 s per kilometer — so re-listening is confined to each flash's predicted arrival window.
- 22 real flash → bang arrivals surfaced from clips the rain noise had blinded — including three flashes ranged in a single 30-second clip:



22
arrivals recovered

20/22
detector recall

0.7 km
median range error

LOCALIZATION

Range without clocks, fixes without over-claiming

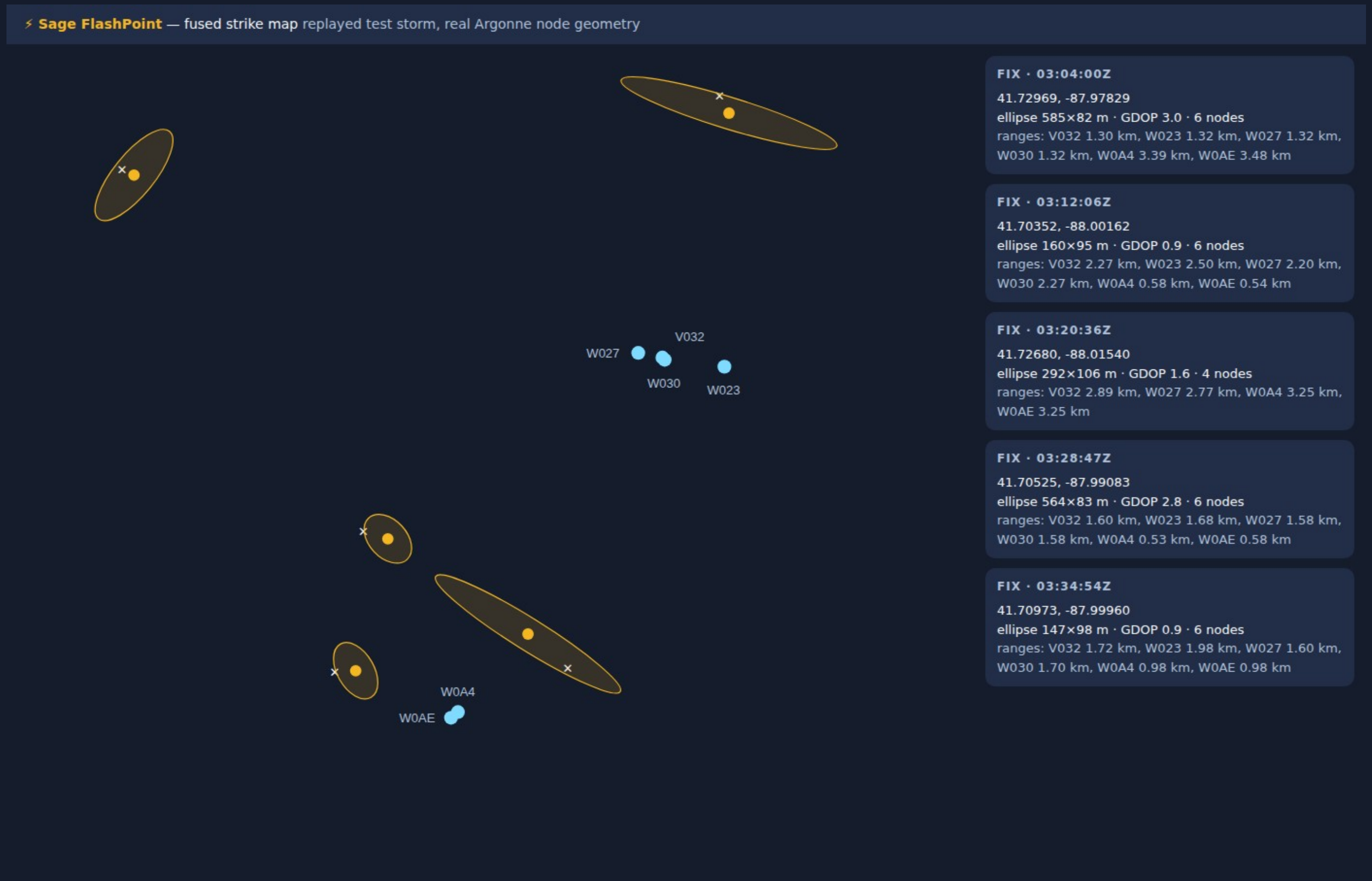
- Light arrives instantly — a camera flash is a free, absolute time-zero at every node. No cross-node clock sync needed.
- range = Δt × 343 m/s, per node; several nodes' ranges triangulate the strike.
- Cross-node consistency (RANSAC — random sample consensus) picks the one candidate set that agrees; a geometry gate labels every solution fix, ambiguous, or range-only — the system never over-claims.

96 m

fix with no satellite, no sync (integration test)

5/5

strikes fixed in the replayed test storm, 70–273 m



Fused strike map: a replayed test storm over the real Argonne node geometry (cameras + microphones only).

THE CONTROLLER

Storm mode armed 145 minutes before the first flash

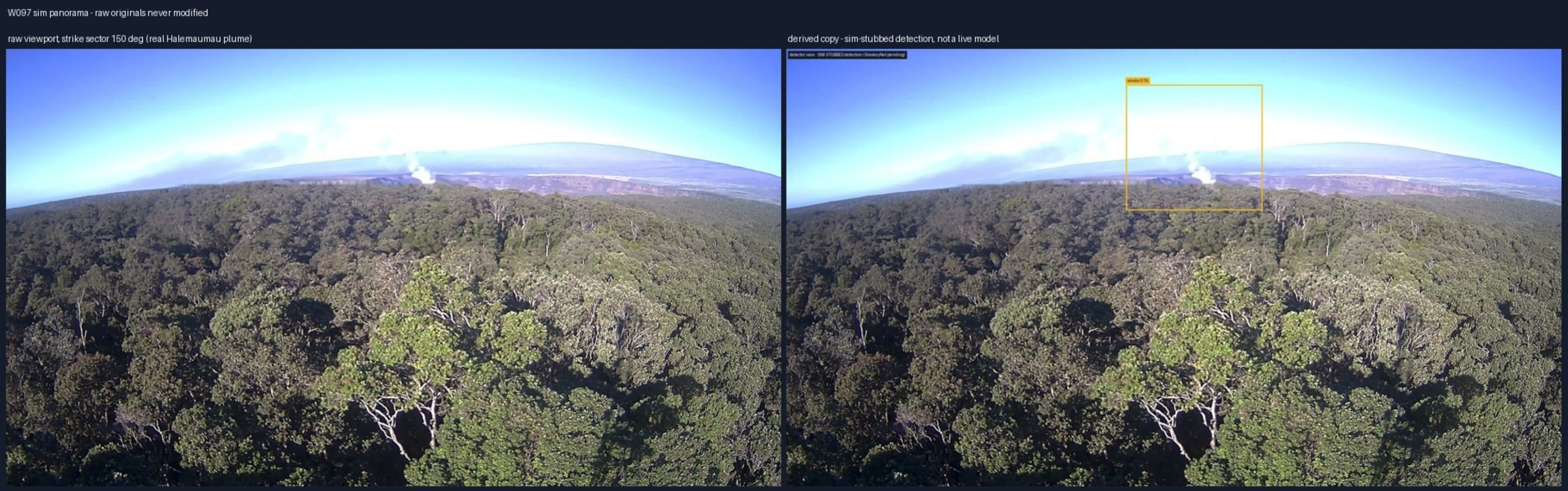
IDLE → OUTLOOK → APPROACH → STORM → AFTERMATH

- Free NWS (National Weather Service) outlooks arm it cheaply; radar cell tracks and local sensors escalate it; every action is budget-capped, hysteresis-damped, and audit-logged.
- **Replayed against the real Kitten storm feeds, it armed 145 minutes before the first local flash — then scheduled the holdover watch and expired cleanly. 19/19 tests.**
- Why it matters → the frame on the right: through the entire ignition storm, the node's steerable camera pointed at a cabin wall.
- Two failures, honestly separated: tasking (fixed by this controller) and siting — W06C sits low in forest, so smoke confirmation needs elevated partner cameras or better-sited nodes. Both remedies are recommended.

W06C's PTZ camera during the ignition storm (top rows) and the falsified events (bottom row).

HOLDOVER WATCH + HUMAN REVIEW

Every strike becomes a 72-hour watchpoint

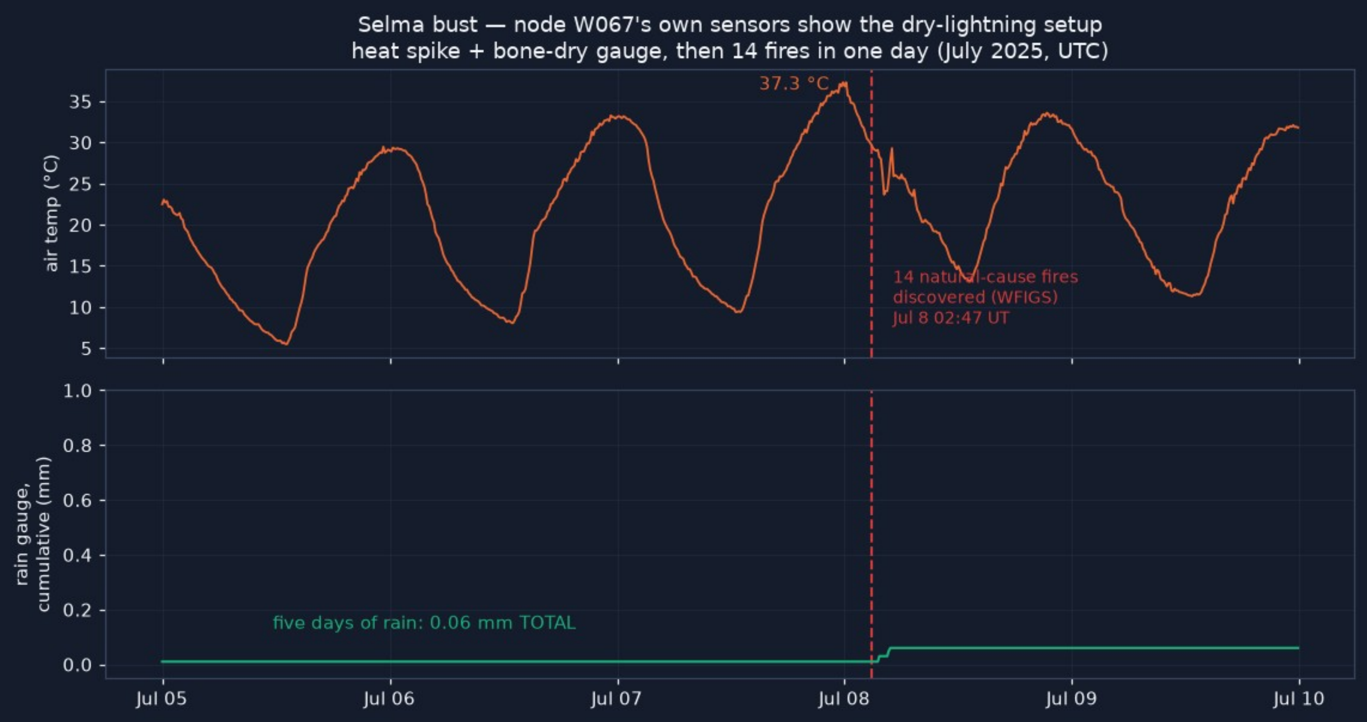


Evidence ships as an untouched raw frame plus a labeled derived copy. The boxed detection here is a test-harness placeholder — in live gateway runs, Gemma 4 captions each frame and YOLO11 draws detection boxes; SmokeyNet is the planned smoke detector (W097 panorama; the plume is a real Halemaʻumaʻu gas plume).

- PTZ patrols strike sectors by risk, dwelling to the smoke model's needs — validated through the real sage-agent gateway on the camp node (test-mode PTZ, live Gemma 4 captions).
- Ignition-risk score with per-factor provenance: node rain gauge, NASA (National Aeronautics and Space Administration) POWER (Prediction of Worldwide Energy Resources) dryness, fuel, detection confidence.
- Alerts land in Slack: one glanceable summary line, evidence strip inline, full forensic audit in-thread.
- Reviewers click Confirm / False positive / Keep watching — decisions flow back to the node AND become labeled training data. No automated dispatch, ever.

THE SECOND CASE

Fourteen fires. Five days. 0.06 millimeters of rain.



Queried live during camp week from W067's public weather streams. Fire records: NIFC (National Interagency Fire Center) WFIGS (Wildland Fire Interagency Geospatial Services), agency-determined cause.

- Selma, Oregon: 14 natural-cause fires logged in ONE day, 14–34 km from node W067.
- The node's own gauge: 0.06 mm across five days, under a 37 °C spike. Its only movement — a 0.05 mm trace — lands exactly on the fire line: the ignition storm itself, essentially dry.
- **Risk score v0.1 back-test — separation is already clean:**

87	Kitten dry night storm → fire next day
84	Selma bust → 14 fires next day
41	wet-storm control → low risk, correct
0	no-strike day → no card, correct

THE HONEST LEDGER

What was built, what was found, and the limitations

BUILT THIS WEEK

- Noise-adaptive thunder + flash detectors (anchored listening)
- Sage/Waggle edge plugin — built and RUN on a Sage node; candidates published to the cloud API (application programming interface)
- Audio classifier probe v0 — YAMNet + logistic regression, AUC (area under the curve) 0.95 vs 0.70 for the DSP (digital signal processing) score, same labels (first neural result)
- Storm-mode controller — 19/19 tests, real-storm replay
- Fusion engine + strike maps with GDOP (geometric dilution of precision) gating
- Risk cards + Slack human-review loop, live end-to-end
- Forensic dashboard used to build all of the above

FOUND FOR SAGE

- No soil / fuel-moisture sensor publishes anywhere on the fleet
- The manifest is not the archive: 97 nodes list microphones, only 64 ever recorded — and W021 (Colorado) records audio with none listed
- The fleet's #2/#3 most fire-exposed recording nodes (V040/V041, Oregon) ran their microphones only in autumn–winter: 0 of their 35 storm days captured
- Camera siting caps the smoke leg — recommend ridge-top “sentinel” placement or cueing partner networks (ALERTWildfire, HPWREN)
- Node liveness is task-level: W096's imagesampler silent 24+ h while met + audio kept flowing
- Case-node file ACLs (access-control lists; W067 et al.) — access requests filed

LIMITATIONS

- One real storm validates the audio chain (Kitten)
- Fusion + flash validated in replay tests — real multi-node captures need storm mode running
- Ground truth (satellite pixels 8–14 km) is coarser than the sub-km claims — Vaisala research-data request submitted Jul 24, not yet approved
- Read access to key fire-country node archives (W067, W084/W06F, W019) was requested but not granted — additional storm/fire cases went unexamined
- Live PTZ control untested: the W0A4 sandbox camera's web interface was reached over an SSH (Secure Shell) tunnel, but control permissions were not granted — re-aim runs in test mode only
- One hackathon week: v0.1 weights, dry-run job control

The fleet already hears the storms.

Sage FlashPoint makes it listen — and watch what happens next.

FUTURE DIRECTIONS

- **NLDN (National Lightning Detection Network) ground truth** research request submitted July 24
- **Neural classifier** probe v0 landed: AUC 0.95 vs 0.70 DSP — next, reviewer-click labels
- **Smoke leg** SmokeyNet plugin on watch cadence
- **RF (radio frequency) time-zero** SDR (software-defined radio) sferic anchor — daytime parity
- **Hawaiʻi variant** wind-trigger controller on HCDP (Hawaiʻi Climate Data Portal) feeds
- **Live PTZ actuation** camera-control credentials on the W0A4 sandbox unblock real re-aim

Data sources: NIFC WFIGS (fire records, agency-determined cause) · NOAA GOES-18/19 GLM via AWS Open Data · NASA POWER & SMAP (Soil Moisture Active Passive), via Earthdata · NWS · Vaisala Xweather · Sage data & storage APIs · Models: YAMNet, YOLO11 (You Only Look Once), Gemma 4 (SmokeyNet planned) · Maps: Leaflet, © OpenStreetMap contributors, © CARTO

code, data, audio & writeup: github.com/scwatson4/sage-summer-camp-2026 → flashpoint/ · thank you

TARGET OPERATING CONCEPT

Flow of state: from quiet skies to a verified alert

Beehive cloud — orchestrator agent

A more capable LLM (large language model) agent runs where Sage already connects every node — it moves nodes between states; the nodes stay deterministic.

Sees every node
telemetry, detections, camera frames, health — via the Beehive pipeline

Sees the outside world
NASA Earthdata · Vaisala Xweather · National Weather Service · GOES satellites

Directs, never dispatches
chooses node states and cameras; every action logged

The orchestrator supervises every state at right and can advance or reset the flow at any time.

1 Quiet watch

Normal snapshot schedules; the orchestrator tracks forecasts and radar for every node's neighborhood.

↓ *weather conditions suggest lightning near a group of nodes*

2 Always-on capture

Cameras, microphones, and the SDR (software-defined radio) lightning sensor record continuously on the affected nodes.

↓ *SDR registers a positive strike AND enough nodes sit within listening range*

3 Strike triangulated

The strike's radio pulse gives every node the same start-of-clock; thunder delays give distances that cross at the strike.

↓ *location fixed — the orchestrator picks the best-placed camera*

4 Smoke watch

A PTZ (pan-tilt-zoom) camera with a horizon view aims at the strike sector; the smoke agent revisits through the holdover window.

↓ *smoke confirmed at the sector · no smoke by window's end → return to 1*

5 Notify emergency personnel

Before-and-after pictures plus the conditions leading up to the strike — soil moisture, rainfall, wind.

SDR sensing is planned hardware — the camera flash is the start-of-clock today. Notifications are human-reviewed before anything reaches responders. Soil moisture currently comes from NASA data; an on-node probe is the planned upgrade.

BACKUP — FOR Q&A

The pattern repeats on a second node

Fixed fire-detection camera, 17:33 UT — plume FLAGGED

Steerable PTZ camera, same day — pointed at canopy

W097, Hawai‘i Volcanoes National Park, 2025-10-05. Left: the node's fixed fire-detection camera flagged the persistent Halema‘uma‘u gas plume — the detection boxes were drawn by the node's production detector, not by this project. Right: the steerable PTZ camera the same day, pointed at canopy.

Camera capability was never the gap — watch tasking is. The same lesson as the cabin wall, on a different node and hazard.